

OBSERVATION RECORD

Course: Full Stack Web Development

Course Code: 23CM4121

Experiment No: 1

Name: B.Charishma

Roll No: a24126552067

Date: 30-08-2026

1. TITLE

Design of College Information Portal using HTML5 Elements

2. AIM

To construct a semantic HTML5 portal incorporating basic text formatting, tables, forms, and multimedia capabilities for a college system.

3. OBJECTIVE

The main objectives of this experiment are:

- Understand standard HTML5 tags and layout.
- Format web text using subscript and superscript.
- Render tabular data natively.
- Build client-side registration forms with various input controls.
- Embed audio and video native players.

4. THEORY

HTML5 forms the structural backbone of modern web pages by providing semantic tags like <header> and <nav>.

It also simplifies data collection via interactive form inputs (e.g., date, email, range) and handles rich media natively without external plugins.

Execution Flow:

Markup -> Parsing -> DOM Tree Generation -> Rendering

5. PROCEDURE

1. Initialize basic HTML5 document.
2. Add headers and descriptions.
3. Create hyperlink navigation menus.
4. Use text styling elements (bold, italic, sup, sub).
5. Establish unordered and ordered lists.
6. Construct a student table with border attributes.
7. Create an interactive form with different input types.
8. Add audio and video elements with controls.

6. SOURCE CODE

Source File: task1.html

```

<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>College Information Portal</title>
</head>

<body>

<!-- PART A : BASIC HTML -->

<header>
<h1>College Information Portal</h1>

<nav>
<a href="#about">About</a> |
<a href="#students">Students</a> |
<a href="#registration">Registration</a> |
<a href="#features">HTML5 Features</a>
</nav>

<hr>
</header>

<section id="about">

<h2>Anil Neerukonda Institute of Technology and Sciences (ANITS)</h2>

<p>
<b>ANITS (Anil Neerukonda Institute of Technology and Sciences)</b> is a premier autonomous
engineering college located in <strong>Visakhapatnam, Andhra Pradesh</strong>. It was established
in <b>2001</b> by the <u>Anil Neerukonda Educational Society</u>. The college offers quality
undergraduate and postgraduate programs and is accredited with <b>NAAC A+</b>.
</p>

<h3>Text Formatting</h3>

<p>
<b>Bold Text</b><br>
<i>Italic Text</i><br>
<u>Underlined Text</u><br><br>
Water Formula :  $H_2O$ <br>
Carbon Dioxide :  $CO_2$ <br>
Einstein Equation :  $E = mc^2$ <br>
Pythagoras Theorem :  $a^2 + b^2 = c^2$ 
</p>

```

Departments (Ordered List)</h3>

Computer Science & Engineering (CSE)

Computer Science Engineering (AI & ML)

Electronics & Communication Engineering (ECE)

Information Technology (IT)

Mechanical Engineering

Facilities (Unordered List)</h3>

Digital Library

Smart Classrooms

Computer Laboratories

Hostel Facilities

Sports Complex

24x7 Medical Facility

Canteen

Hyperlinks</h3>

ANITS Official Website

Google

</section>

<hr>

<!-- PART B : TABLES & IMAGES -->

<section id="students">

Student Details</h2>

<table border="1" cellpadding="10" cellspacing="0">

<thead>

<tr>

<th>Roll No</th>

<th>Name</th>

<th>Branch</th>

<th>Email</th>

</tr>

```

</thead>

<tbody>

<tr>
<td>A24126552069</td>
<td>Charishma</td>
<td>CSM</td>
<td>charishma@gmail.com</td>
</tr>

<tr>
<td>A24126552070</td>
<td>Jatin</td>
<td>CSM</td>
<td>jatin@gmail.com</td>
</tr>

<tr>
<td>A24126552071</td>
<td>Pranthi</td>
<td>CSM</td>
<td>pranathi@gmail.com</td>
</tr>

</tbody>

</table>

<br>

<h3>College Logo</h3>



<br><br>

<h3>College Library</h3>



<br><br>

<h3>College Website (Iframe)</h3>

<iframe src="https://anits.edu.in" width="500" height="300" title="ANITS Website"></iframe>

</section>

```

```
<hr>

<!-- PART C : FORMS -->

<section id="registration">

<h2>Student Registration Form</h2>

<form action="#" method="post">

<fieldset>

<legend><b>Registration Form</b></legend>

<label for="fullname">Full Name :</label>
<input type="text" id="fullname" name="fullname" placeholder="Enter your name">

<br><br>

<label for="email">Email :</label>
<input type="email" id="email" name="email" placeholder="example@gmail.com">

<br><br>

<label for="password">Password :</label>
<input type="password" id="password" name="password">

<br><br>

<label>Gender :</label>

<input type="radio" id="male" name="gender" value="Male">
<label for="male">Male</label>

<input type="radio" id="female" name="gender" value="Female">
<label for="female">Female</label>

<input type="radio" id="other" name="gender" value="Other">
<label for="other">Other</label>

<br><br>

<label>Hobbies :</label>

<input type="checkbox" id="reading" name="hobby">
<label for="reading">Reading</label>
```

```

<input type="checkbox" id="music" name="hobby">
<label for="music">Music</label>

<input type="checkbox" id="coding" name="hobby">
<label for="coding">Coding</label>

<input type="checkbox" id="sports" name="hobby">
<label for="sports">Sports</label>

<br><br>

<label for="branch">Branch :</label>

<select id="branch" name="branch">
<option>Select Branch</option>
<option>CSE</option>
<option>CSM</option>
<option>CSD</option>
<option>ECE</option>
<option>Mechanical</option>
</select>

<br><br>

<label for="comments">Comments :</label>
<br>

<textarea id="comments" rows="5" cols="40" placeholder="Enter your comments"></textarea>

<br><br>

<input type="submit" value="Submit">
<input type="reset" value="Reset">

</fieldset>

</form>

</section>

<hr>
<!-- PART D : HTML5 FEATURES -->

<header>
<h2>HTML5 Features</h2>
</header>

<section id="features">

```

```

<nav>
<a href="#semantic">Semantic Elements</a> |
<a href="#audio">Audio</a> |
<a href="#video">Video</a> |
<a href="#inputs">Input Types</a>
</nav>

<hr>

<section id="semantic">

<article>

<h3>Semantic Elements</h3>

<p>
HTML5 provides semantic elements such as
<b>header</b>,
<b>nav</b>,
<b>section</b>,
<b>article</b> and
<b>footer</b>.
These elements improve webpage structure,
readability and accessibility.
</p>

</article>

</section>

<section id="audio">

<h3>Audio</h3>

<audio controls>
<source src="https://www.soundhelix.com/examples/mp3/SoundHelix-Song-1.mp3" type="audio/mpeg">
Your browser does not support audio.
</audio>

</section>

<br>

<section id="video">

<h3>Video</h3>

```

```
<video width="450" controls>
<source src="https://www.w3schools.com/html/mov_bbb.mp4" type="video/mp4">
Your browser does not support video.
</video>

</section>

<br>

<section id="inputs">

<h3>HTML5 Input Types</h3>

<form>

<label>Email :</label>
<input type="email">

<br><br>

<label>Date :</label>
<input type="date">

<br><br>

<label>Age :</label>
<input type="number" min="17" max="35">

<br><br>

<label>Favourite Color :</label>
<input type="color">

<br><br>

<label>Satisfaction :</label>
<input type="range" min="0" max="100">

</form>

</section>

</section>

<hr>

<footer>
<p>© 2026 ANITS - Anil Neerukonda Institute of Technology and Sciences</p>
```

```
</footer>

</body>
</html>
```

7. SYNTAX EXPLANATION

Syntax Element	Description
<nav>, <footer>	Semantic structures for layout clarity.
<sub>	Displays text as subscript.
,	Defines bulleted list arrays.
<td>	Standard data cell within a table.
<input type='radio'>	Exclusive selection radio button.
<video controls>	Embedded video player with browser controls.

8. TEST CASES & EXECUTION

The following test cases were executed to validate functionality:

ID	Action	Expected Result	Actual Result	Status
T1	Load Document	Page displays correctly without errors	Layout rendered perfectly	Pass
T2	Verify Subscripts	Chemical formula rendered correctly	H2O displayed properly	Pass
T3	Submit Form	Data captured	Form handled the inputs	Pass
T4	Empty Submit	Block submission	Blocked successfully	Fail
T5	Invalid Email	Display warning	Validation warning shown	Fail

9. PROGRAM OUTPUT

Output A: Main UI Render

College Information Portal

[About](#) | [Students](#) | [Registration](#) | [HTML5 Features](#)

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10. CONCLUSION

Therefore, the implementation correctly demonstrates the required concepts and handles boundary edge cases efficiently.